

ActInf GuestStream #016.1 ~ Mark Solms

"Consciousness as Precision Optimization"

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all right hello everyone welcome to actin flab guest stream number 16.1 with mark solms it's january 22nd 2022. it's gonna be an awesome discussion we really appreciate everyone taking the time out to participate in the live stream and in the discussion we're going to have some intermittent presentation intervals by professor solms and uh also a period for discussion alternating with that so we'll each just introduce ourselves and then pass it directly to the presentation section so i'm daniel i'm a researcher in california and i'll pass to dave i'm dave i'm a software developer and student of psychology i live 200 kilometers north of manila hello i'm stephen i'm based in toronto canada and i'm doing a practice-based phd around social topographies and how people engage in community development and i'll pass it over to mark now welcome thanks very much so what i propose to do is to in a moment i'll share my screen and you'll see my slides um i'm going to start with the summary slide of all the points that i propose to cover and then i'll take them one by one each of those points and at the end of each of them i'll stop and see if anyone wants to comment or ask a question and then we slowly plow through it like that we've got two hours so it should be okay but if worse comes to worst and we don't get to the end of the list of topics we will arrange a follow-up meeting to conclude to conclude them so i'm very glad to be here thanks for inviting me and i'm really looking forward to this discussion here come my slides um on the screen is the title of my book and here is the promised summary of what i'm going to cover as i said uh i'll i'll cover each of these points in turn and uh if we don't and then we'll have questions and comments in between and if we don't get to the end that's fine let's rather let's rather give each one of the topics the time that it needs so i start with the with the um assertion that affect is the foundational form of consciousness and that affect is intrinsically conscious i then move to make the point that that claim that affect is intrinsically conscious in other words that you can't have a feeling that you don't feel that this is not based on semantic arguments but rather on empirical evidence and i will summarize the evidence i'll then make the point that affect is not

synonymous with interceptive inference which is widely believed these days and i'll tell you why that matters when i get there and then i'll move on to making the point that all of this casts new light on the hard problem particularly if affect is not just an intraceptive form of perception because there are lots of problems about perception that have got us into this heart problem in the first place i'll argue that affect rather than being intraceptive inference that it's an extended form of homeostasis i'll tell you what that what i mean by that when we get there and i'll make the point that complex organisms like ourselves require multiple homeostats and that these are to be treated as categorical variables which matters and that these variables must be prioritized one can't meet all one's needs simultaneously i'll then uh this is an a large point this one is a large point the the the empirical evidence about affect and this one is another large point the mechanism of perceptual and cognitive consciousness is precision modulation of allostatic prediction errors that's a big claim and so we must spend some time on it i'll then talk about the nature of the predictive hierarchy i seem to see it a little differently from most people so i want to set out my stall on that score and explain why i think the predictive hierarchy is arranged concentrically and that this with as one moves toward the periphery of the hierarchy there's progressively increasing tolerance of uncertainty and then lastly i'll say a word or two about why i think that consciousness can be artificially engineered so even if we don't get to the end of all of these topics one way or another even if we're the second meeting those are the topics we will cover and we'll see how far we get tonight and at least you know what the over what the overall when i say tonight i'm in berlin i know it's morning for some of you but at least you get the overall picture from that slide so let's start with affect is the foundational form of consciousness and it's intrinsically uh the discoverer of the unconscious i quote him uh because freud more than anybody wanted us to realize just how much of mental life is unconscious even freud said that it is surely of the essence of an emotion that we should be aware of it that is that it should become known to thus the possibility of the attribute of unconsciousness would be completely excluded as far as emotions feelings and affects are concerned it's quite a startling statement from as i said the discoverer of the unconscious to say that the possibility of the attribute of unconsciousness would be completely excluded as far as emotions feelings and affects are concerned now i know this is a controversial issue and that there are all sorts of ways in which people want to argue that there are such things as unconscious emotions and what not um and this has partly to do with the fact that different people mean different things by these words uh emotions affects and so on um and it has there are also all kinds of methodological um operationalizations of these things uh i won't go into

all of that i'm happy to if any of you want to discuss it but i've i've emphasized the word feelings to make clear that if anyone wants to argue that they think that the word emotion can apply to unconscious processes or affects can apply to unconscious processes that's fine by me but the word feelings i don't see how you can ever argue that feelings can be it just doesn't make any sense that a feeling could exist without being felt if it were not felt it would not be a feeling and so i'm going to focus everything that i'm going to say on this process this this thing this this mental state called feelings and i'm arguing that feelings are the foundational form of consciousness and that these this form of consciousness is intrinsically that's as a sort of a dogmatic statement i'm not sure if anyone wants to discuss perhaps you want to hold your horses but that's the end of my first point so anyone who wants to interrupt here go ahead otherwise i'll move on thanks for the digestible claim-based format my question would be what else has been posited in your view as a foundational form of consciousness so what are the counter facts for that first claim what other horses are in the race the reason i focus on effort um well i shouldn't say the reason i focus on effort because that's not quite what you're asking you're asking what are the other horses in the race so let me say that the whole search for a neural correlate of consciousness which is in a way where the trouble began with with francis crick it was focused on visual perception and they're good reasons to focus on visual perception uh mainly because if you take a look at your you'll excuse the pun take a look at your consciousness right now it's probably dominated by visual perception uh visual qualia predominates in our consciousness a very large part of our cortex is given over to visual processing everybody's always believed that cortex is where it's at when it comes to consciousness and we also understand a very great deal about visual cortical mechanisms and visual mechanisms all together so that was a perfectly reasonable place to start so that's the other main horse that's in the race but vision was thought to be a good model example not because it is the totality of consciousness but because it was assumed that if we can crack the problem of how visual consciousness arises then we can generalize our conclusions to other forms and i'm going to argue that there's something profoundly different about visual consciousness and affect of and moreover that there's something profoundly different about cortical consciousness and the brain stem mechanisms which is where affect of consciousness arises and i know that i'm saying that boldly now and i'm about to argue that point uh in the second of the of the bullet points uh on the screen uh affect of consciousness is not a cortical form of consciousness it's a brainstem form and that the cortical form of consciousness is contingent upon the brain stem form this is this is one of the main reasons apart from

evolutionary reasons which sort of interdigitate with and flow from that from that anatomical localization um why i think that affect is the foundational form so it's really vision that i'm using as my account my kind of foil to answer the question simply stephen one question with feelings is would you say that feelings are integrated with proprioception in the sense that they um are felt sense of where the body is as well so it brings it into kind of an action-orientated domain or is it enough to have um sort of a sense of how things are going sort of an overview of just an emotional state and then whether those two connecting some way as i'm going to argue when i get to my third point i don't think that affect is synonymous with intrceptive inference altogether interceptive perception altogether um and i don't know some people would would class proprioception with interreception uh some people would not some people would say it's a sort of a it's a sort of a hybrid between intercepted and extrceptive perception but to the extent that by the word proprioception you're referring to a form of perception i would disagree with that because i don't think that affect is a form of perception i think that proprioception follows um if if by proprioception you mean what what neurologists normally mean by that word it's it's a it's it's part of the broader category of somatic sensation therefore uh uh that does not uh qualify uh i think as affective because as i say i'm going to argue that affect is not a form of perception there there i draw a distinction uh as do you know i think it's it's perfectly uh um fine to draw a distinction between extrceptive bodily sensations and interceptive bodily sensations but i think that the interceptive type is is is nothing other than perception and not intrinsically different so even if you think of proprioception as straddling those two domains i still think that it's a modality of perception and suffers from all the same problems as vision for that reason i see we have a comment in the chat line i'm not sure how how we're going to deal with that if one of yous i'll read this it was a question in the live chat from ingrid they wrote what differentiates affect from feeling from your perspective is affect conscious or pre-conscious thank you um you know this is uh a perennial problem we don't all mean the same thing by these terms so that's why i've focused on the word feeling because there can be little doubt what feeling denotes it denotes something you feel in other words something that you are necessarily aware of if you were not aware of it you it wouldn't be called a feeling because you wouldn't feel it affects and emotions denote many different things to different people when you ask me is affect pre-conscious or conscious and so on it depends what you mean by affect um to my mind and i can only say um you know just make clear how i use these words uh i use the word feeling to denote the subjective experience of affect uh affect is a kind of abstract term for a function that we feel um but that doesn't mean that you

all have to use the word in the same way and it's precisely in order to not get lost in this kind of semantic tangle uh that i'm focusing on the word and i'll make clear in the very point that i'm moving to next uh what i mean by feeling in terms of its um it's its physiological and anatomical and other objective instantiations excellent stephen you you have your hand up probably just didn't oh is that your previous hand so just go to the second point okay very good so the second point is as you saw we've just been talking about the meaning of the words um and i've i've um staked my claim for the word feeling to make clear that that's what i mean when i say that affect is intrinsically conscious i mean that it is necessarily felt um i'm also saying it's the foundational form of consciousness that doesn't flow from the word except perhaps from the word in the sense that the way that nagel and chalmers define consciousness and many people seem to have sort of agreed with this way of defining it that there is something it is like to be conscious that that this something it is likeness it seems to be more or less the same thing as feeling it seems to mean the same thing but feeling in a rather broader sense of the word than affective feeling so i want to be clear now why other than those other than those linguistic or semantic points why i'm making this claim about feeling it's not by any means based in the semantics that you have to feel a feeling i i want to be clear why i think that this is the foundational form of we've we've all known since when there was this big surprise that magoon and mirzi found that all consciousness is contingent upon what we subsequently called reticular activating arousal in other words upon the something that's contributed by these core brain stem structures which after magoon and maritz's work came to be known as the reticular activating system i want to be clear i'm not only talking about the reticular activating system but also about the periaqueductal gray which is not technically part of the reticular activating system but it is part of the brain structures which seem to be fundamental to the generating of consciousness um the what magoon and marutzi showed among other things what was that if you make a small lesion above the reticular activating system thereby severing it disconnecting it from the forebrain and from the cortex in particular then consciousness is obliterated entirely consciousness is 100 dependent upon brain stem arousal that was a shock and a shock that i don't think we have fully come to terms with even all these years later it should have led us to realize that we've been looking in the wrong place uh for the neural correlate of consciousness if we're looking to cortex because it's plain even from just that study performed in 1948 that cortical consciousness is derived from elsewhere cortical conscious consciousness cannot occur in the absence of reticular arousal but the way that magoon and maruzi fudged it was to say and everyone more or less in their wake has said

the same that it turns out they're two kinds of there's the brainstem kind uh which we call wakefulness or the level of consciousness um and then there's the cortical kind which we call awareness or the phenomenal um contents of consciousness so this distinction between a quantitative level contributed by the brain stem and qualitative contents contributed by the cortex the way in which we save the cortical theory of consciousness following magoon and lewis's seminal observations in case you're wondering what this refers to uh it refers to a more recent study by fisher and colleagues which showed that the smallest lesion necessary to produce coma in human beings is a two cubic millimeter lesion in other words a lesion the size of a match head in the parabrachial complex so there's no doubt on the basis of magoon and meruzi studies all those years ago and on the basis of these much more recent observations that consciousness cannot exist without whatever it is that this part of the brain contributes this this primitive brainstem call but this idea that it's merely a a level or a or a quantitative dimension of consciousness is what i i think is unsustainable and i'm going to very quickly because i'm sure many of you have heard me talking about this before i going to very quickly whiz through the evidence for why this this notion that the brain stem contributes only a quantitative level a sort of a wakeful prerequisite uh for consciousness uh rather than equality and the content um but i want to before i do that to just give you a clear picture of what is meant by this distinction um the the the distinction uh is is is analogous to uh the difference between a television set and its power supply uh the idea that the cortex is responsible for the contents and qualities of consciousness for phenomenal consciousness the real stuff the real failure of consciousness where the thing that all the arguments are um it's that this is like the television set and it has to be plugged in uh of course you can't a television set can't do its television thing unless it's powered up uh to claim therefore that the real source of television and and all that that word represents uh is the power supply the electrical uh source uh in the wall uh is ludicrous and and that's pretty much how we've all seen until very recently how we've all seen what the brain stem contributes to consciousness so nobody's doubting that it's necessary that it's prerequisite uh that that without it you can't have cortical consciousness but it's been it's been uh reduced to a mere power supply something which is not actually consciousness itself in the sense that that we're all arguing so why do i think that that's not correct well uh i'll start with the most dramatic evidence which is patients who have no cortex if the theory that the cortex is necessary for the contents and qualities of consciousness then patients without cortex should be in a coma or at the very least if we claim that the brain stem produces nothing other than the sort of background level patients with no cortex should have

some kind of blank wakefulness i suppose the closest uh natural kind of thing that tallies with blank wakefulness is the vegetative state which is also called non-responsive wakefulness in other words there's wakefulness without anybody being there without it there being anything it is like organism that has blank wakefulness uh due to the absence of cortex but when you look at cases who who have no cortex and here's the scan of a girl with hydrant encephali born with no and you examine the girl herself you see that she's not only conscious in other words not in a coma she's conscious in the sense that she wakes up in the morning goes to sleep at night but more interestingly she is responsive remember my definition or the definition of the vegetative state the state of blank wakefulness it is non-responsive wakefulness and clearly this girl is not non-responsive here she's responding to the placement of her baby brother on her lap and she responds emotionally in a way that any kid any little girl place a baby on her lap she'll she'll she'll most will express pleasure most most kids like their little baby siblings being placed on their laps um and it's not just her it's most of these kids if they're if they're raised in something other than a state of complete deprivation like romanian orphans who fall into what might look like a although their cortex is entirely intact if you actually raise these kids in in reasonably humane conditions they are conscious and responsive uh affectively responsive that's the main point that they're making these children with absolutely no no cortex their consciousness has a quality and the content that quality we call affect it feels like something to be these kids and the content is not monotonous affect it's all kinds of effort that he has a summary slide for extracting what beyond marco who studied many many many of these kids what he said about them they show all of these different emotions and most importantly of all they show them in situationally appropriate ways in other words they respond with laughter if they're tickled uh they respond uh with with arched backs and and and and and protests uh if something that they want uh is denied them uh they respond with a startle uh if there's a sudden uh unexpected noise and so on uh the evidence is that these kids are not only conscious in the sense of being awake which is interesting in itself in fact i can't imagine what a state of blank wakefulness might be like but there you have it the the the the the concept uh derived from lagoon and naruto's work is that there is such a theoretical fiction as blank wakefulness but these kids are not blank uh these kids are emotionally responsive in exactly the way that you would expect if they had consciousness at least in the form of affect so i'm arguing that with no cortex at all display the content and qualitative conscious contents and qualities in the form of affects now of course the big methodological problem is that how do you know what it's like to be these kids how do you know if there's anything it's like to be these kids they can't tell us anything

about what it's like to be them so uh all you can do um in this situation is look at other lines of evidence if we can find multiple converging lines of evidence converging on the conclusion that the brain stem is where the affects are actually generated we can do no better we can do no more we can do nothing other than that in science so i want to show you now uh what happens if we look at cases you can speak who've lost great swathes of cortex we they can't have lost all of their cortex because otherwise they wouldn't be able to speak to us so i'm wanting to show you examples of cases who have lost great swathes of cortex and the cortex in question is the cortex that has been that that has been most closely identified uh with with sentient being with with the subjective presence of a mind of a conscious mind the first example is of prefrontal cortex i think that it's fair to say that the mainstream view is that where all of this information processing comes together and is received by a sentient subject of the mind the mainstream view is that would be in the prefrontal cortex now there are few patients there it's rare to have a patient with absolutely no prefrontal cortex but here's one example and i'm just giving you examples in in this presentation i'm speaking about individual cases those individual hadron and cephalic kids that you just saw um they are representative of the class of patient that i'm talking about here is a patient of my own um who has no prefrontal cortex but happily uh has a sliver of language cortex and so he's able to tell us what it's like to be him i asked him if he was conscious of his thoughts and he said yes of course i am and i explained to him that according to classical theory he shouldn't be in order to confirm that he's conscious of his thoughts i asked him if uh if he would be willing to solve a problem that would require him to consciously picture a situation in his mind and he agreed to do that so i said the i want him to picture in his mind i want him to imagine that he has two dogs and one chicken which is which he then says okay he'll do and then i ask him do you see them in your mind's eye and he says yes he does i then ask him to count the legs you know this is not an everyday task it's something that's unexpected so there he has these two dogs in his mind's eye and a chicken and he has to count the legs and of course the answer is 10. i ask him how many legs do you see and he says to my great disappointment he says eight i say eight and he says yes the dog's at the chicken uh which um i think is uh enough evidence for me that this chap uh is there you know that there's there's there's somebody at home uh who's looking at these dogs and the chicken and imagining perhaps concretely you know that the dog's actually at the chicken or perhaps he's pulling a fast one he's making it a joke and in fact the making of pure oil jokes is a very common feature of patients with the frontal lobe syndrome and this is linked traditionally with the commonplace observation that these patients are emotionally

disinhibited and why i draw your attention to that is that if it were true that the cortex and the prefrontal cortex in particular is where affects are generated then how come if you have an absence of prefrontal cortex not only is affect preserved but affect is excessive uh there's there's more effect than normal there's there's too much effect this is entirely incompatible with the idea espoused by many respectable neuroscientists joseph le Douarin for example who's a great authority on affect joseph le Douarin says that of course the amygdala and other subcortical machinery for anxiety doesn't actually generate any feeling at all that the feeling of anxiety is read out to use his term labeled by the prefrontal cortex and that's when it actually turns into a conscious experience so i say again if that were true then how come not only my patient this one making this joke to us uh but all frontal lobe patients the the less frontal cortex they have the more affect they have so the less capacity they have to read out and label their feelings somehow it seems as if they're beset by more feeling rather than less um since i'm talking about feelings the other major candidate the other major cortical candidate as being the seat of the sentient self is the insula bud Craig claims that the insular or perhaps even just the anterior third of it is where the sentient subject comes into being and he links this particularly with affect he says that this is the part of the cortex which registers the state of the own body how do you feel now is registered by this part of the cortex and it's on this basis that he claims that this is where the sentient subject comes into being so again in just one case like marzia who had absolutely no insula bilaterally obliterated and he interviewed in much the same way as i interviewed my patient with no frontal cortex and demario asks him if he has a sense of self and he says yes he does and i won't go through all of this all of which is demario trying to ascertain whether or not this person is conscious and in particular whether or not he possesses selfhood does he even understand what self food is and eventually demario says what if i were to tell you that i know you better than you know yourself patience says i would think you're wrong demario says what if i were to tell you that you're aware that i am aware patient says i'd say you're right the marzia says you're aware that i'm aware patient says i am aware that you're aware that i'm aware please note all these references to i uh there is a sentient subject speaking to us here insular cortex so i think the claim on the basis of the hydrant and cephalic children the claim that the cortex is the seat of consciousness and of feeling in particular has to be dismissed uh but they can't speak so we look at the other major candidates who can speak patients with no prefrontal cortex they are conscious there's a sentient subject there and they feel like something they tell us and they show us and likewise patients with no insula of course the claim can be made well in these cases their consciousness arises from the

remnants of the cortex the remainder of the cortex the quarter the cortex that is not removed bear in mind the ones who have no cortex you know who clearly show behavioral displays of affect every prediction that you might make from the hypothesis that it feels like something to be these kids is confirmed by their behavior by their responses to affective stimuli in that case we say well we don't know because they can't speak so then we use these other cases with no prefrontal cortex no insular cortex who can speak and then we say well it must be the the other bits of the cortex that are generating the consciousness it seems like a losing wicket to me let's look at other evidence then lesion remember we're looking for converging lines of evidence converging on the conclusion that the feelings are generated in the brainstem in the brainstem and so here we look at the case of a patient in where an entirely different method is used that is deep brain stimulation of a part of the reticular activating system in this case the substantia the surgeon was aiming for the subthalamic nucleus but the the electrode went into the substantia and when he stimulated it the patient fell into a profound depression within five seconds again i'm not going to read all of this to you it'll it will take more time than is necessary if you read it yourself you'll see this patient within five seconds felt fell into a profound depression as she has no psychiatric history never been depressed before never had a diagnosis of any kind led to learn a diagnosis of major depression and here within five seconds you can generate produce cause a suicidal depressive state which within 30 seconds of switching off uh the uh the the stimulator 90 seconds i mean after the stimulation will stop the depression stops very interestingly she agreed or very bravely she agreed to have the stimulator switched on and off either in the subthalamic nucleus or in the substantia of the reticular activating system where she was blind as to where the stimulation was being applied and every time it was applied in the reticular activating system she fell into this severe depressive state again so this is you know this is what you would expect this is the the prediction uh from the hypothesis that ethics are actually generated in the brain stem the the the prediction would be if you stimulate these reticular activating nuclei you will generate affects and that's exactly what you do this is also why i mentioned the periaqueductal gray earlier because if you stimulate reticular activating nuclei you generate intense effects but the same applies to periaqueductal gray in fact the greatest intensity and range of affects anywhere in the brain can be elicited from deep brain stimulation of the periaqueductal brain brain stimulation of the cortical mantle produces very little if any affect um another method positron emission tomography so another study of demarzio's patients in states of anger sadness happiness and fear uh you see where the brain the major brain activity is in all of these affective states the major activities subcortical it's in

the brain stem it's not cortical so again this is exactly what you would predict if the affect is actually being generated by the brain stem not being read out by the cortex but generated by the brain stem and then there's also the matter of the neurotransmitters that psychopharmacologists tinker with the mainstream psychiatric medications modulate neurotransmitters that are sourced in the reticular activating system we don't normally think of it that way but it is simply the case that serotonin which we increase with antidepressants is sourced in the reticular activating system dopamine which we block with antipsychotics system noradrenaline which is blocked in many anti-anxiety drugs system if the reticular activating system was just the power source uh that i referred to in my analogy about the television set then it should only interest anesthetists you shouldn't be able to change the channel alter the programming of the television set by pulling out the plug or by or by attenuating the power supply you might expect the screen to sort of go blank or or just flicker but not to have whole new programs with new contents and new qualities and so this is my argument that the the idea that feeling states affective feeling states of the kind that those children display of the kind that these patients describe of the kind that psychiatrists are or directly trying to manipulate are generated by upper brain stem structures they're not generated by cortex you don't need to have cortex in order to have these states and yet we all know and agree we all agree that cortical consciousness is contingent upon reticular activating arousal if we now are forced to accept that reticular activating arousal is not contentless is not a mere quantity it's not just a level but rather has content and quality that we call affect then we have to accept that affect is a prerequisite for all other forms of consciousness and it's on this basis not on semantic one that i'm claiming that affect is the foundational form of consciousness uh it also makes good evolutionary sense uh that this ancient structure rather than the cortical mantle which evolves only with mammals that the ability to register your own subjective state how you're doing effectively in terms of hunger and thirst and sleepiness and to hotness and whatnot that this should not require such a late evolutionary development but that's another point i'm just wanting to slot in there that i also mean it in the evolutionary sense when i say that consciousness i don't only mean it in the functional sense now i'll pause there because that's a big claim and a very controversial claim in fact i must tell you i'm still slightly dismayed that it remains controversial i've been i've been really surprised by the degree of pushback that i get when i argue and it's i'm not the only one who argues that i'm not the first one to argue it demarzia argues that panxip was the first to argue it marker argues it here i come a good few years on and argue the same thing citing all of this evidence and uh wow

people just don't want to accept it so let's hear what what what you have to say yeah yeah i'll read a question from the chat first and then we'll go to stephen uh jonathan in the in the chat wrote i can imagine a modern large language model the likes of gpt3 creating such conversations without apparently having a mind's eye while i agree it is good evidence i think we are seeing machine learning systems having similar conversations so you've presented case studies that are the tip of the iceberg of broader populations and analyses how do we think our evaluation of conversational dialogue in questions about necessity and sufficiency here yeah you see this is a kind of a um a relative of the philosophical zombie uh argument uh the the problem is with this sort of thing is that in fact the same applies uh to you and me i can't know for sure that daniel is conscious that stephen is conscious that david is conscious or that the person who asked that question is conscious you know the point you've just made really does apply in exactly the same way there's no greater reason to believe that it applies to the patients that i've just presented to you then and that it applies to you and me because you too could be a philosophical zombie in other words you too could look as if and you have internal states say that you have internal states but actually there's nobody there i can never disprove that that is a metaphysical problem that's not a scientific problem the scientific method is requires that your hypothesis has to be consistent with all the known facts um and then you have to make falsifiable predictions from that hypothesis and if your predictions are not falsified then they are upheld unless uh and until uh they some evidence is found which positively this is what positivism is all about which positively falsifies it so i'm saying these kids look as if they're conscious you know they behave as if they're conscious now it's true that they might not be and stephen might not be and daniel might not be and david might not be but you know my hypothesis is that they are so what do i do i say well if daniel feels pain like i do then when i stick a pin uh in his hand he's going to go ouch and withdraw his hand and avoid me in future and that's uh you know that's a confirmation of my prediction uh the same applies for these kids but you know that's just one source of evidence as i said you don't accept just one you then say okay well what about this line of evidence what about that and i've given you five lines of evidence here in which in every instance uh the prediction from the hypothesis that these people are conscious and that the consciousness is generated in the brain stem and that that it has an affect of quality every single line of you know the prediction is confirmed what you would expect would happen if that were true is what happens i'm afraid you can do no more than that in science um and it's a curious fact that we raise the bar uh of evidence uh when it comes to the problem of consciousness um i think we must be very careful not to

require um any different sort of evidence in the case of consciousness then we require for any other matter in science and in this case in any other biological function you know it's if you start with the assumption that consciousness is something special that you know the ordinary rules of science don't apply to then you're going to have a self-fulfilling prophecy so yes it's possible it's possible that these kids and those patients are not there that they're lying to me uh that it's it's a little bit of a curious thing that somebody without a sense of self uh wants to pretend that they've got one um i think you're in a kind of slippery slope there to to uh to irrationality i can't prevent you from holding that view it's just not consistent with any of the evidence i'm afraid thank you mark stephen yeah i was wondering how much you think the challenge with uptake is because people are sort of stuck in the idea that knowledge and information is a signal coming in meaning it's all about meaning and the ability to communicate meaning as opposed to engaging in meaningful action and skillful inactive processes of the voice twitching the voice box twitching and enabling words to be in the space around us in some sort of way and whether that that issue is um kind of a deeper ingrained challenge to overcome well i completely agree with that i think that that is the a fundamental source of the problem because our common sense tells us our consciousness is extraceptive as i said earlier you know the our conscious experience is dominated by visual qualia and where do they come from from you know things light waves impinging on my retina from outside and i'm seeing things out there and so consciousness seems to be so obviously derived from something out there um and this is this was at the heart of the experiments that magoon and maruzi did magoon and maruzi weren't expecting to find what they found um that they were expecting that a cat deprived of any extraceptive stimuli will fall asleep um if not fall into a coma and it was therefore a great surprise to find that that's not what happens that the consciousness is generated endogenously comes from deep within the brain doesn't flow in with sensory stimuli um and when they lesion that core brain stem structure the lights went out they couldn't make sense of it and they came up with this with this uh theory which as i've shown you at least as i've shown to anyone who's you know who's willing to look at the evidence and draw the conclusions uh that uh this that the the way that they framed it is wrong it doesn't explain what they saw what they saw was that consciousness is endogenously generated and where does that where does the notion come from this common sense notion that it flows in from outside it comes not only from our own everyday experience but also from the british empiricist philosophers who probably themselves derived it from common sense you know that the mind literally the mind as a whole you know is is derived from

sensory experience that leaves impressions uh leaves memory images impressed upon our cortex which are then associated with each other and this according to british empiricist philosophy which underpinned the whole of 19th century and early 20th century neurology and neuropsychology it's all predicated on that idea so i think it's deeply ingrained within ourselves and within our science um and uh you know within perception generally that consciousness comes from outside and you you're speaking about some further um you know ramifications of that and therefore sources of resistance against the alternative view but all the evidence suggests that consciousness is endogenous consciousness is generated from within i mean you saw i i had i i i'm sorry i don't know who the person was who made the comment about you know how do we know that this is not you know just some sort of program spewing out language um i have nothing against you i don't know who you are i just hear that all the time and uh you know i i find that frankly i find it astonishing you know that you present all of this evidence and then still people say well you know how do you know but how do you know that that these patients are really conscious and this is a this this this demonstrates the extent of the of the prejudice you know that has come with that with that historical um with that legacy stephen with a raised hands then dave with a raised hand we'll go with the raised hands okay thanks um yeah so sort of following on from that yeah i totally agree and actually i think the work with first piston and active vision um is is quite helpful because the idea that we're reaching out and feeling the world and i think actually i was looking back before 300 400 years ago if you went back in time people actually almost had that as a an intuition that we were looking out in the world but once we discovered the retina and then photography we kind of re-imagined so one question i've got is is regarding cerebral palsy i work a lot with communities with complex disability with non-verbal and their pre-motor cortex is often damaged and in the 1970s it was thought medically if you couldn't speak you couldn't think so that was where they were then if you couldn't communicate words and then once they started using symbols bliss symbolics and the ability to then and they suddenly discovered that all these clients who had who were non-verbal actually perform complete sentences because they have a way to act in the world and interact in the world and develop um language syntax and all of that so i'm just thinking whether you've had a lot of uh whether any of this works been tied in with clients with cerebral palsy not only with cerebral palsy but with all sorts of neurological conditions if i can just sort of piggyback on what you've said about cerebral palsy the whole trend within clinical neurology has been us realizing that patients who we thought were not conscious actually are and the whole the the sort of tenor of the research that's been

done in recent decades um i mean locked in syndrome of course is the extreme example but but i don't only mean locked in syndrome i mean all of these states of reduced and obtunded uh con and clouded consciousness and so on minimally conscious states so-called vegetative states and so on we're finding more and more patients who actually who we thought were not conscious with increasing technology we find that they are with with improvements in our methods we find that they are so it's it's a sort of an odd um it's an odd um contradiction with the sorts of philosophical skepticism and doubt that one confronts from people who say well you know you know how do we know they're they're behaving as if they're conscious but maybe they're not in fact what we've been finding in clinical neurology is patients who behaving as if they're not conscious and then when we when we use the proper tools we find that they actually are your comment about cerebral palsy can be multiplied you're the observations that you refer to consciousness um you know using reportability as a criterion the capacity to declare your conscious states has really it's not just led to methodological and epistemological problems it's really led to gross clinical errors you know it's ethical problems i i feel we need a lot more than mere philosophical skepticism before we can take seriously in neurology which after all is a branch of medicine before we can take seriously the possibility that these kids and these patients look as if they conscious behave as if they're conscious say that they're conscious and we say well we can't be sure i mean think about the implications thanks mark dave with a question yes um two i'd like to say two things uh to what you were just talking about uh the difficulty of telling whether people who seem to be locked in or comatose are there have been some people who initially got their experience in working with the diet they just wanted to people who were dying to not be alone and they spent a lot of time as typically as volunteers sitting with folks who seem to be comatose who seem to be completely unresponsive and they found that events that unroll very very slowly in the patient's room seemed to elicit a response and they got in high-speed cameras and would uh record what the patient is doing that's a study that's also been done with with infants that infants seem to to act extremely fast uh and that if you speed up those images you can actually see a child responding who seems to be kind of lying there they found with the comatose that it seemed they can carry out conversations and it actually did resemble conversations taking a person through childhood scenes for instance and watching them respond and they nod or they'd shake or they grin you mentioned some kid that used to bully you over a period of minutes maybe saying it over and over or speaking very slowly and you get this very slow but eventually a full grin that a nurse would say oh that's just a grimace now let me check her

diaper and the other thing is a much better bigger picture question who actually is hyper alert getting out of professional settings or ideological settings you walk around you see someone who's just you got to be really careful with and they will just startle you have to be really careful what what you say you don't bring up unpleasant topics you never imply that they've done something wrong who are these for the most part these hyper alert individuals are children of alcoholics who as small children had to be the responsible person they had to keep daddy from hurting mommy even and i wonder how is niels bohr raised was he hyper alert is that why he became amazingly oppressive as nice as he might have been nonetheless unremittingly insistent that only he was so in that big picture thing are the folks who want to insist that you absolutely positively are controlled only from the outside were they the child the people who as children were never able to emit they always had to react they always had to be really really careful because the environment really is something that will come in and get you well um about your second uh comment i will just say uh please note uh how affective uh you all of this stuff is that you're talking about and i i i think that we we really underestimate the importance of affect in terms of the whole gamut of phenomena that we call consciousness but to your first point i want to make clear uh that i am i mean i agree with what you're saying uh the about the the inference of consciousness in these uh but i but i want to make clear that i'm making actually a quite a in light of or notwithstanding or you know all of this other evidence and all of these other very interesting uh clinical phenomena all of which uh point in the direction of we've underestimated the presence of consciousness you know that we've we've been relying on a very poor criteria in terms of reportability and so on that really we need more sophisticated methods and the more we use more sophisticated methods the more we find evidence that there is in fact somebody i agree with all of that but i'm making a very very limited claim i'm saying i'm not saying these kids are like you and me i'm not saying they have a full rich inner mental life i'm saying only that there is something it is like to be them um and in light of all the evidence in their cases plus all the other evidence that i've cited or using all these different methods all i'm saying is that that's something it is likeness that these kids possess is it feels like something they have feelings that's all i'm saying so it's it's all very interesting all these other things about how all these other capacities um have been missed by us in relation to all of these other neurological conditions but in these cases which have no cortex whatsoever they have only some of them have basal ganglia and and you know other subcortical for brand nupi some of them don't but even in cases you have no forebrain only a brain stem they display affects and so i'm saying that the brain stem prerequisite form of

consciousness does have a quality and the content it therefore feels like something there is something it is like to be these kids and and that quality and content is affected that's all i'm claiming that the foundational form of consciousness is affect and it's generated in the brainstem it's a it's a it's actually i think a rather modest claim in light of all the evidence it's quite a conservative claim it's a sort of it's a sort of a deflationary claim but so much flows from just that and i i'll just say once more i find it very hard to understand i mean i can understand you know and of course or politically or what you know i mean people's whole careers i mean generations of careers have been built on the assumption you know that that that cortex and consciousness um are are intrinsically bound up with each other which you know so the people have a hell of a lot to lose and and many of us with egg on our faces and i freely admit that i uh too uh believed that the cortex was the organ of consciousness this is what i took in like mother's milk when i trained and i believed that for many many years after i trained you know so you know i'm not exempted from that but gosh we've got to look at the evidence it's wrong we made a mistake you know it's not doesn't mean the cortex does nothing it doesn't mean it doesn't contribute anything to consciousness it's just not the source of consciousness the the cortex is elaborates this very raw primary elemental feeling that is generated in the brain stem and all we need to do is accept that and uh and then we'll make huge progress in consciousness science uh the the the the fallacy that the cortex is the source of consciousness uh is really a serious impediment to us at this point uh in the history of our field i i think it's you next daniel yes it is indeed a theme of the discussion that ambiguous evidence are continually interpreted and reinterpreted in light of our inherited or our previously structured priors and that's true in the active vision example it's also true in the scientific example where our inherited priors about scientific models influence how new evidence is not just perceived but funded and carried out and um [Music] it made me think about uh when people activate a gene or remove one gene and make claims of function might be an analogous to the brain but removing or having one node in a complex network showing that there is a single point of failure or gain of function does not entail that as a generating mechanism which is just an example of an explanation that retains an alternative perspective and so i think it's interesting on multiple levels as we kind of apply this idea of updating and learning and structural learning on our theories about ostensibly the part of the body that does this function as opposed to one's hand which is important and has nerves in it but doesn't do this kind of engagement thanks very much um so i agree that's an everyday observation um that uh you know we we are reluctant to accept prediction errors uh we we we cling to our

priors especially our most deeply consolidated priors we see this in everyday life nobody likes to find they were wrong that they have to change their mind that they have to accept evidence for review other than the one that they had always held and this applies even to our scientists i mean who uh in their graduate studies when they were doing their first research you know didn't hope uh that their hypothesis was right uh and then when the evidence came in and they found oh dear you know i haven't found what i expected to find it's not welcome so what i'm referring to based on what you've just said um i know this is not quite what you were saying i'm sort of elaborating what you were saying um is that we don't recognize uh sufficiently the affect of dimension uh in our resistance um to prediction errors you know we we we we all know about you know priors and posteriors and you know and how this all works mechanistically but just from the sort of anecdotes that i've mentioned um i'm i'm i'm i'm trying to get us to recognize that there is an affective dimension to that and when we get to this point about the concentric predictive hierarchy i'll come back to but i think it's an it's an it's an interesting and important observation that as you traverse the hierarchy um and and get to what uh in my way of thinking about it as a concentric hierarchy to get deeper into the hierarchy in the traditional way we speak of going higher in the hierarchy but as we go deeper further from the sensory motor periphery we come upon priors which are which are more and more difficult to update um and uh i say again there's a there's a there's a feeling that goes with that which is i don't care what you tell me i'm not gonna believe you thank you mark stephen with a question yeah i know you're going to come back to the cognitive consciousness um but just something around terminology we've been doing work around an ontology working group at the active inference lab actually trying to understand how to think about the language and you mentioned it could be um affects it is generated in the brainstem and then and that's when something's like that's what it's like but i'm also wondering how are you taking the dynamic with the body and environment going down i don't know if that's seen as another manifestation because obviously it makes sense because it's it's more integrated with the rest of the body's nervous system it's less distant from it but um i'm wondering whether the term um generated within is more to do with when it's connected to the rest of the brain and when it goes down it's actually more distributed in the mind body environment dynamic i would have nothing against i mean that way of thinking is um i'm very sympathetic with it i'm i'm not a narrow localizationist uh i'm i'm very um very happy to recognize um it comes naturally to me to think about functional systems which are distributed and dynamic and and in fact the the reticular activating system is probably the most deeply interconnected with everything else in

the brain system that there is um it is it is it has tentacles everywhere and not only although it's called the ascending reticular activating system it descends also you know it's uh it modulates the whole nervous system um in very important ways that that will come to you know because this whole thing of precision modulation um is at the heart of my way of understanding um so the the main thing that i'm wanting to say is i agree with you uh that the the this is a deeply uh interconnected a system which functions uh by modulating other systems i would be very happy if it were if we had to leave it there and say well you can't quite say that affect is generated in the reticular activating system because you know it's it's really modulating these other systems my reluctance however in this case to accept that um is twofold uh the one is that that's the kind of thing that people like joe ladue are saying they're saying yeah this is all unconscious until it is labeled by or read out in uh you're not talking about cortex uh so you know that's but the principle remains you know that this i i i don't want to concede the point that it only becomes conscious once it modulates something else and that that's really where the contents and qualities are that's a sort of theoretical reluctance on my part you can see why but more importantly and more tellingly uh it is the evidence uh when you look at these kids they have no cortex so their efforts can't be modulating it can't be the result of something that the brainstem is doing to cortex because there ain't no cortex and yet you know there's all the evidence that they have affects um and so i think uh as much as surprising as it may be uh you know it's it's it's not consistent with our normal way of thinking about cortex and it's also not something that i'm wedded to the idea that that that you know anything is generated in anything but the evidence suggests that certainly uh the modulatory influence that the reticular activating system has over the cortex true as it is important as it is fundamental as it is to how that system works that the affects are not a product of the modulation of cortex the effects are generated in the subcortical structures and if you just take a step back and the the implication of that is that well maybe there are affects in vertebrates who don't have cortex in other words maybe affects are not the exclusive preserve of mammals um and actually that seems pretty likely it seems pretty darn likely non-mammalian vertebrates have aspects and again you know when you start looking at the evidence it's pretty darn good that they do so um i i'm happy to accept everything that you're saying about let us not forget that this is a system which is deeply interconnected with other systems that its primary function is to modulate other systems but all i'm saying is that when it comes to cortex it's not that the affects themselves are not generated by the modulating of cortex that doesn't mean that there's a disembodied brain or disembodied reticular activating system but you can

have a decorticated reticular activating system and it can generate effects nonetheless thank you mark dave do you wanna thanks go okay so one thought i see that the the comments in the chat line have they been dealt with can i move on let's try to get to at least one other of these important points oh well this is a brief point um you know the reason i'm making this um about affect not being synonymous with with interceptive inference is because those few people uh working within the active inference framework who are not corticocentric and happily they are increasing you know there's there's uh people who worked within this framework they had no reason to question you know all of the mainstream assumptions of cognitive neuroscience as far as as far as cortex is concerned um the the the but slowly um it's become apparent uh that affect uh is is uh is an important topic that we we're not only talking about perception and action and cognition uh you know we're talking also about affective processes and and about you know body regulating processes and so on um in fact the whole of self-organization uh you know has to do with the regulation of the of the own body and you know the whole notion of self-evidencing systems and so on you so so this is happily slowly coming and increasingly coming uh into the mainstream of um the af people working within the within the sort of active inference framework um but the the the dominant view within that uh school of thought is that affect is to the internal body uh what perception is to the external world in other words extraception uh is uh it is a perceptual inference regarding uh things going on outside of and affect uh is perceptual inference about things going on inside of me um and that's you know if that were so that would be nice and simple um but i i don't see it that way so uh and the the perhaps the outstanding proponent of the view that affect is intrceptive inference is a neil seth and he's just brought out a a really very good book um i have it right here here it is being you um you know which which is predicated on on on that view so because it's because i agree with him about so many things i i think that it's easy for people to conflate our views and to imagine that what i'm saying is the same as what he's saying on this score so i want to make clear that i'm not saying the same as what he's saying on the score that why do i think that affect is not the same as perception uh it's but but rather interceptive perception well first of all because there are many interceptive perceptual states which are not effective so you know you can you can perceive this the the your heart rate uh the beating of your heart you can perceive uh your your respiration rate or the or the the the the the inspiring and expiring uh of air from your lungs you can perceive gurgling uh in your tummy which is an unequivocally visceral and all of these things you can perceive them indifferently you can perceive them without any affective valence or quality at all so if affect

were synonymous with interceptive inference then why are these common and garden examples of interception not effective secondly if affects were intrareceptive perception how come so many affects are generated by extrareceptive events where there doesn't seem to be any clear interceptive driving force so for example pain which is a fairly simple example of an affect you know you stick a pin in my finger it comes from the outside and it penetrates the you know the nociceptors in my extrareceptive nervous system i feel this intense affect called pain so that's extrareception and yet it's effective um the same applies to surprise you know due to fright you know you get a you get a you get a you get a shock um well in fact electric shock is another one but these are but i i mean just you know fright something unexpected happens outside of you and you startle and there's a there's a there's an affect that goes with that a particular type of disgust is another one you imbibe something um that tastes smells bad and you wrutch it up with this terrible feeling of disgust and then they're more and they're not purely entirely non-bodily affective states like missing somebody you know missing your brother it's clearly affective and there's nothing bodily about it and nothing intercepted about it when i say nothing bodily i mean particularly you know it's not it's not about the state of your intrareceptive visceral processes so i'm saying there are interceptive visceral processes which we perceive which are not effective and they are affective processes uh affective states that we experience which are not interceptive so i don't think that we can say that affect is interceptive inference that it's the same as external perception except it's directed toward the state the internal state of the own body i think that it's something else and i don't say this to be difficult you know i'm i'm actually quite uh concerned that those few of us within the active inference framework who are recognizing the importance of affect uh can't agree on this point because uh you they're few enough of us you know who recognize the the importance of africa so i'm i'm just wanting to make clear why i think that it's not merely a kind of mirror image reflection um of of extrareceptive uh uh inference and then i want to add one last point which is that when you read i mean everything that i'm going to go on to argue about why extrareceptive perception remember what i said at the very beginning about cortical vision having been the model example of where we will find the neural correlative consciousness which has led us down this blind alley and and got us got us into such uh such a mess that that it has to do largely as i'm going to argue later with the fact that cortical processes and perceptual processes are not intrinsically conscious things you can perceive the world through all your extrareceptive modalities vision included at the cortical level too you can perceive the world without perceiving it consciously in other words extrareceptive inference perceptual inference is not an

intrinsically conscious process there's nothing about extraceptive inference or perceptual inference that that requires consciousness to for us to understand the mechanisms at work and so if interceptive inference is just the same thing applied to the interior of the organism it suffers all the same that it's it's not clear why it becomes conscious um and i said that's going to be my last point on the school but in fact i realized there's one more that i want to make which is that um the vast vast vast majority of our intraceptive processes are unconscious um i mean even more so than our extraceptive processes most of our regulation of our visceral bodily economy happens autonomically we know nothing about it so um it would be again a very curious place to find uh the um this the essential mechanism of of consciousness and in this case of affect if the majority of the mechanisms of this kind in fact do not operate consciously and if it suffers the very same problems our extraceptive um using using vision as a as a model example using any form of perception i think as a model example of consciousness is to make a mistake because consciousness is not an afferent process consciousness is an endogenous process it's something generated not from within the guts but from within the core of the brain and uh that that's where we're going next so i'll stop there for any any uh comments on that point before i move to the next one great presentation important distinctions and points so indeed we'll take a breath i think we can head on to the next one okay thanks so now i've told you in relation to the point i've just made i i think that uh the hard problem uh of consciousness has arisen uh in large part from the example that we've used of consciousness starting with crick who has to be credited with bringing consciousness into center stage uh so you know i'm not i'm not bashing crick um and i also have said already i think it's perfectly reasonable that he started where he did um you know it's clear why he started with visual perception as his example where we should look for the neural correlate of visual consciousness and then distinguish you know unconscious vision from conscious vision and then there we have it then we have the neural corumat and then we can generalize from there um i'm saying that it turns out that was an unfortunate place to start and it links with what i was saying earlier that's also why i'm saying i don't want us to make the same mistake just just just turning it inwards and saying we're dealing with perceptual so let me let me say why all of this casts new lights on the hard problem um yeah so yes chalmers the person who formulated the hard problem in the very paper that he formulated it and he says what makes the hard problem hard and almost unique is that it goes beyond problems about the performance of functions to see this note that even when we've explained the performance of all the cognitive and behavioral functions in the vicinity of experience they may still remain a

further unanswered question why is the performance of these functions accompanied by experience a simple explanation of the functions leaves this question open why doesn't all this information processing go on in the dark free of any inner field so this is a core aspect of chalmers's hard problem he's saying that our normal way of going about doing cognitive neuroscience where we take a mental phenomenon and we say okay uh you know let's see uh you know which part of the brain needs to be damaged in order to disrupt this this mental uh state and then you know let's start piecing together what the what this part of the brain does and how it works and uh next thing we end up with a wonderfully complicated map of all of the mechanisms at work in in this case using the standard example thomas's point is that you can do this this is standard the standard approach in cognitive neuroscience but it doesn't tell you why there is something it is like to see you can you know the famous knowledge argument with mary the visual neuroscientist she knows everything that there is to know about the functional mechanisms of visual perception but she knows nothing about what it's like to see in in the in the in the example that jackson or the story as jackson told us you know she's never seen color but you know i i like to extend and simplify his arguments and just say she was born blind she knows everything there is to know about the functional mechanisms of vision but she's never experienced vision and if one day she's blessed with the gift of sight she will learn something utterly new about vision namely what it is like to see what redness is like what cleanness is like and so on and this is their argument the argument is the normal uh way we proceed in cognitive neuroscience namely to dissect the function uh and in this way to explain uh the mental thing that we're trying to explain works very well for all of these mental functions but it doesn't work very well it doesn't work at all for defined as what it is like to perform these functions uh they're claiming chalmers is and nagel and others of that ilk that somehow the consciousness is in a different realm uh from every other mental function because you know when you when you've explained all the functional mechanisms you still are left with this epistemic gap between that explanation of the function and the qualitative experiential phenomenal feel of seeing so i'm saying that the problem is that they're using perception as an example perception is not an intrinsically conscious function and indeed cortical perception is not intrinsically conscious that it is demonstrably the case that the cortex can perform uniquely cortical perceptual operations actually recognizing color you can recognize color unconsciously and you can recognize faces unconsciously you can read words unconsciously so um if the cortex can do all of this perceptual and cognitive gymnastics without consciousness i think it's perfectly reasonable to say well why doesn't all this

information processing go on in the dark without any inner feel because that kind of information processing can go on in the dark without any inner feel a perceptual inference is not intrinsically conscious this uh statement of chalmers is this really a strong statement where he says there is no cognitive function such that we can say in advance that explanation of that function will automatically explain experience this is this is what he says on the basis of everything that i've just told you this is the this is this is where he's led he's saying that they you that explaining the function can never explain experience if that's the case we're in big trouble because consciousness is somehow you know going to have to be dealt with in a different way from everything else in science but if you change that word cognitive to affective um traumas would charmers have said this there is no affective function such that we can say in advance that explanation of that function will automatically explain i'm saying of course there is the function of feeling i am saying in advance that explanation of the function of feeling will automatically explain why it feels like something because that is the function of feeling the function of feeling is for the organism to feel it that to feel its own state so i'm saying had we begun there had we begun our quest uh for the neural correlate of consciousness with this model example had we had we begun by asking why and how does it come about feelings are experienced i think that we would have had a much less hard time with it than we have had with starting with with visual information processing the mechanism of the mechanism even of cortical vision is not intrinsically conscious it does not automatically explain experience because it does not automatically have to be experienced but the mechanism of feeling you cannot explain how feeling works why feeling exists what it's there for if you have a functional account of feeling it would have to account for why it feels like something in other words why it's experienced so i think that what i said in my earlier point that um that um yeah that these facts uh uh cast new light on the hard problem uh the this is the new light that that it casts i just have also uh on the screen this slide this famous review paper by killstrom which summarizes all the experimental the conclusion that we were led to actually a good four decades ago already that perception without awareness of what is perceived is perfectly commonplace and learning without awareness of what is learnt is perfectly cognitive and perceptual functions performed by the cortex are not intrinsically conscious processes so this question of chalmers is why is the performance of these functions accompanied by experience why doesn't the dark that perplexity is not difficult to understand in light of those facts but i don't think that this question has anything like the same gravity uh when you apply it to the kinds of mechanisms that give rise to feeling i'm saying in short that

if mary was an affective neuroscientist rather than a cognitive neuroscientist or a visual neuroscientist she could not know everything that there is to know about the functional underpinning feeling without expecting that it will generate something that is experienced without without the mechanism necessarily accounting for why it is experienced and this is why i think at shifting our focus from cortex to brain stem and from perception to affectum has big implications for our ability to finally come to grips with this hard problem of consciousness i'll pause there again thank you thank you mark i'll read a comment in the chat and then you can respond then stephen so dean writes pre-arousal plus awareness minimum of two not as brain geography or territory establishment effort rather a dance in an evolving relationship as we might see in an agent dancing with the environment whereas the territorial view might examine control brain stem of the cortex or vice versa one in an in controlling relationship through a mechanical lens this relationship of arousal and awareness could be choreographed but not choreographed mechanized but not necessarily choreographed not sure what that part meant our feelings a residue of arousal plus awareness revealed we accept the signals like we dance out a path generated look i think that this is similar to the point that we were discussing earlier you know about um generated in uh you know the upper brain stem as opposed to the upper brain stem modulating other other other structures whether they be above or below the brain stem in the i said then i'm very happy to think of it as a dance uh as something choreographed where it's not a matter of whose master and whose servant you know um that that's a is it's a when it comes to the relationship between the affective arousal um and and cortical representation i think that there's some instances where affective arousal just absolutely um brushes aside any amount of uh decision making and thought processing and conversely uh the opposite happens too you know that there clearly are ways in which where one can in a top-down way modulate affect dramatically and i don't only mean you know regulate your emotions i mean if you look at some of these examples of these buddhist monks who are able to you know really massively control autonomic functions and and the affects that normally would go you know with uh not breathing or walking on uh you know painful um surfaces and so on that those affects seem to be overridden so i have nothing against the idea of there being a choreography between brain stem and cortex i've already said why i think that in this instance the affects do not require the cortex actually qualify for the name affect that they actually are felt without cortex i've already said why what the evidence is both in terms of studies of uranus studies using the various methods that i mentioned the most dramatic being as i said cases where there is no cortex and yet there's evidence that there's

there's plenty of affect there but also the evolutionary implications of that which is that we don't have to wait for the appearance of cortex and the mammalian series for uh that for the dawn of consciousness that if if if affect is generated in the subcortical structures uh then um affect uh it does not require cortex and affect is a it can be um can be uh attributed other vertebrates and not only vertebrates by the way there also is good evidence that some invertebrates are conscious of the octopus is the famous the most famous example um but i think that there's a further problem in this comment you know which has to do with the word awareness um you know you speak of arousal and awareness and you seem to attribute arousal to particular activating system which i'm happy with and awareness to cortex which i'm not so happy with if if if you mean that you cannot be aware of arousal without cortex so i think this word awareness it's like all the other words i mentioned at the beginning like emotion and affection and i said i want to i want to use the word feeling to make clear what i'm talking about because these words mean different things to different people the word awareness means different things to different people and to to many people it seems to mean something like reflective awareness in other words you know in other words being aware of what you're aware of or access consciousness as Ned Block calls it you know that there's this sort of phenomenal consciousness which is some kind of like a raw consciousness and then there's a then there's a subject that that accesses this information it's it's like the global workspace theory works that i mean it's pretty similar sort of idea um if if we're going to reserve the word awareness for that higher order thought or reflective consciousness or access then it means that we can't attribute um awareness to raw fields uh and um that's that's an odd use of the word awareness to me um to me how can you feel pain without being aware of it you know and and pain um you don't need to have any kind of reflective cognition about it you don't need to have any explanatory labeling uh or anything it's just like that's so and you pull away even if you don't have language even if you're not human even if you're not a mammal even if you're another kind of creature so um i arousal for me implies uh awareness um in the way i use the word but um if if we're going to use awareness for that higher order type of consciousness then i would just want to remind you that there are other types of that don't possess that kind of awareness which nevertheless have something it is likeness um and and the core of that quality is feeling that something it is like this is yes thank you um so i was wondering if we think of um consciousness as a kind of an integration of our action expectations of what's happening those expectations then were and Casper Hesp talks about this with his work with active imprints but is how those expectations were weren't met can be effective um so i was

wondering within that kind of that prediction process um when someone gets a question right so you have a mathematician when they get it right they get a feeling that it's right it's like it's a felt sense and you could ask someone what's that feeling like and they can sort of bring out a metaphor landscape that relates to that like it can actually have a structure and there's a similar thing with pain as well pain has to be expressed as a metaphor you have to say it's like a stabbing or it's a throbbing or it's a it's always some kind of dynamical kind of metaphorical embodiment so i'm just curious um has these you know this embodiment this meeting in this brainstem uh how you how you how you see that in terms of if that's congruent with you and whether this this is then uh sort of a re-reimagining consciousness is like a reimagining of of these expectations um and it can it could go up or down in terms of how that is being kind of realized so i'd like to separate out two different aspects of what you're saying the the first has to do with metaphor i think that may be true of um of being able to report or declare your conscious states in language uh that somehow or other you have to symbolize because language is is just that it's a symbolic representation thing itself that as long as you you use the word expressing it yes you can't express in words and emotion without using some sort of metaphor some sort of symbolic system in order to be able to communicate it to others they can of course be the direct behavioral effect of display expression of the emotions in darwin's sense of the use of that term uh that which we then use symbolic species um you know we we use that symbolically uh too but um i i don't think that the raw feeling itself has to be re-represented in any way i think it can just be the raw feeling um without it being represented in in in any language in the broader sense of the word and so i don't think that metaphor a necessary component of the core feature of feeling then the other thing you were saying which had to do with uh with surprise uh uh if i understood you correctly um i i i at times in my own thinking about this there were times when i was keen on thinking about affect or linking affect more with error signals than with than with inferences themselves um the and that was you know i was thinking along those lines because so many people within the active inference framework were speaking about perception you know this famous phrase controlled hallucination that perception is fantasy that it is the it is the upshot of the predictive process um and uh what worried me about that is the very thing i've been banging on about all along you know which is that perception is not intrinsically conscious and so this business of inferring the source of your sensory signals doesn't explain why there's something it is like to infer the the source uh uh of your or the cause uh of your of your sensory signals and so so so from that sort of point of view um i've i found myself thinking well it it's not so much that you

know i mean imagine if what you perceive is just what you predict that that that would be you know bloody dangerous um so the controlled aspect seemed to me to be the more important aspect it's the it's the constraint on what you predict um that seems to be more the upshot of what we call perception as as opposed to hallucination and so i i wanted to place the emphasis um more uh on on the error side of things than on the than on the prediction side of things but subsequently i i have changed my mind uh on on that score um for which will become clearer you know when i when i get down here where is it uh when i get down here um which we're obviously not going to get to today uh but i'm i'm very much looking forward to us having a second meeting where we can get there for me uh it's it's got everything to do with the uh with the modulation of confidence in the prediction uh versus uh the area signal or the error signal that is that that is connected with uh that particular prediction um and and i will just say it now because we're not going to get to it in this meeting um that that the pithy formulation of the way that i see it is that it is good when things turn out as expected and it is bad when uncertainty prevails and i mean here goodness and badness um in as in the in the evaluative sense of the word that is linked to the primary um value system of all living things you know which is that it is bad to die and it is good to survive and if things are turning out as expected in other words you have a prediction which actually cuts it that that is good from the viewpoint of the organism um and that if things are not turning out as expected and uncertainty is uh prevailing that is bad uh it means you know this this is likely to end in tears and so um the uh what was it the choreography that somebody with the dance that somebody else was talking about earlier and i think that what that that the the crucial thing here is the is the the palpating of the precision uh the modulating of the precision uh in the predictions uh uh over the error signals uh and that that is at the heart of what affects all about and i'm only saying that now uh in advance because i know we're not going to get there in in this meeting because we've only got seven minutes to go but i'm going to try if i may just unless there any further questions and comments are there any further questions and comments no here are some future comments it takes many oh they are so then if they're hostile something i think what we should do is i'll take i'll take whatever comments remain or questions and then we'll start next time with this point because i don't think i can squeeze it in to i don't think i should even try to squeeze it into to too short at a time period so totally agree yeah that was my comment was uh we will have a last few minutes really just appreciate you coming on for the 16.1 and we'll find the time to do 16.2 when we will condense and sort of reflect on what has already transpired and look forward to these next points i think it was really great to walk

through it this way stephen or dave do you have any last thoughts otherwise mark please have a last thought so dave first then optionally stephen and then mark yeah in addition to professor seth whose book we already have who else can we uh check with as people doing ai without the um the constriction with without which you mean with you mean who else is working on affective processes within within this exam um two people who come to mind immediately is manos sakiris both great and also both of them are very good speakers uh so so if you if you're wanting to invite them on to uh this this um seminar what live stream series i i think that they would be um they would do well awesome thank you very much stephen yeah just to say thanks very much and um yeah we've been reading some of the work of ryan smith and casper hesp so oh well ryan smith also orion and and maxwell ramstead they're also working on affect they both are they i should have mentioned them ryan smith and and maxwell ramstead sorry that they're doing some fabulous work i really loved their recent paper on folk psychology uh it was just brilliant excellent oh thanks well we'll we'll be connecting those dots up i'm sure next time as well thank you okay well um thoughts yes mark yeah just just to say that i appreciate having the um i mean it's not every day that you have two hours uh with with with excellent minds uh to be able to talk about only half of what you wanted to talk about so it's been very kind of you uh to um to um give me this this this latitude uh and i think that these issues and i'm also particularly pleased that you're going to be in one way or another recording and i don't just mean video recording but also turning into some sort of written form uh what we're talking about because as i hope you can see from what i've said just in half of what i wanted to go through is really are pretty basic issues for for consciousness science and for the science of the mind and for and by implication uh for what we can do within this active inference framework in tackling these really i don't mean to be hyperbolic when i say these profound and ancient problems i think that we have real prospects of being able to make significant progress um on on these fronts and so it deserves is the point i'm making it deserves the time that that we are giving it and i'm grateful to you for um for recognizing that well sir thank you very much thanks thank you thank you